

Beyond the Paycheck: An Empirical Assessment of Worker Economic Well-being in a Changing Labor Market

Abstract

In an environment where the current landscape remains in a period of market volatility, fluctuating in a delicate balance between prosperity and poverty, the social repercussions acting on the U.S. workforce are commonly overlooked. Although more wealth is commonly associated with a higher regard of socioeconomic standing, this view point fails to acknowledge external factors that contribute to this measure. To fully evaluate the scope of economic security amongst these laborers, this research paper answers the prompt: What factors contribute to economic well-being among U.S. workers, and how do employment conditions influence financial resilience? Through a quantitative analysis of the 2025 Survey Household Economics and Decision Making administered by the Federal Reserve and a review of over 18 research papers, I developed a quantifiable Economic Security Index composed of five dimensions of individual weighings: (a) Income Adequacy, (b) Access to Benefits, (c) Job Flexibility, (d) Income Stability, and (e) Financial Resilience. With these factors identified, I composed a computational model¹ to generate visualizations measuring the relationships among these variables, enabling a comprehensive assessment of the current economic standing of citizens across the United States. The findings reveal that economic security cannot be accurately measured by income alone, instead emerging from the intersection of these five measurable dimensions. Of these dimensions, stability proves to be the most influential, regardless of raw income itself. These insights suggest that addressing labor force vulnerability requires a shift away from variable or gig-style compensation towards guaranteed, stable income frameworks.

¹ Github Repository containing source code, processed data, and documentation:
<https://github.com/RishikaBasnet/Beyond-the-Paycheck-An-Empirical-Assessment-of-Worker-Economic-Well-being-in-a-Changing-Labor-Market>

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Introduction

Economic security is a fundamental component of individual well-being, playing a vital role in a person's ability to meet basic needs, withstand financial difficulties, and schedule for the future accordingly. As our economic environment and labor systems around us remain in constant flux, shifting accompanying employment structures, it becomes increasingly difficult to measure economic success and security. Traditional indicators such as annual income or employment status provide rudimentary information, but often fail to capture the broader financial circumstances that dictate whether individuals are truly economically secure. Access to workplace sponsored benefits, consistent income, financial resilience, stability in income, and income adequacy all contribute to an individual's financial stability, yet fall short from being examined as a contributor to the larger system.

Existing research has addressed each of these dimensions separately, commonly investigating the effects of employment benefits on short-term financial security, the role of income volatility in household financial stability, and the importance of emergency savings in protecting individuals from unexpected economic downturns. On a similar note, studies concerning workplace flexibility have proven that greater autonomy can correlate to an improved work-life balance and overall job satisfaction, whilst papers in income adequacy prove correlations with the popularized nuclear family systems throughout the United States. However, limited research exists that combines all these factors into a single quantitative framework capable of calculating cumulative economic security. Consequently, policymakers and government officials often rely on isolated indicators that may omit the interactions between multiple aspects of financial well-being.

To account for this gap, this study develops a multidisciplinary Economic Security Index (ESI) that combines the aforesaid dimensions of economic welfare: Income adequacy, Access to Benefits, Job Flexibility, Income Stability, and Financial Resilience. Rather than evaluating these

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factors independently, the proposed framework aggregates them into a standardized index to represent an individual's authentic economic security. The index is constructed using data from the 2025 Survey of Household Economics and Decision Making administered by the Federal Reserve, accompanied by evidence from over eighteen peer-reviewed studies to guide variable selection.

To evaluate this proposed blueprint, this study creates a computational model that transforms survey responses into standardized scores, calculating an Economic Security Index for each participant and analyzing the relationships among its five variables using visualizations. Accordingly, this study addresses the following research question: What factors contribute to economic well-being among U.S. workers, and how do employment conditions influence financial resilience? The ultimate answer to this prompt provides insights into which dimensions contribute most strongly to overall economic security, as well as manifest how computational methods can support future economic research and evidence-based policymaking.

Literature Review

To fully grasp what drives economic well-being among U.S. workers requires more beyond tracking wages, it requires a refined definition of what makes a career secure enough to protect a household against financial shocks. This review draws on six sources from the Quality of Work Life (QWL) literature as a means to justify five parameters used to construct the Economic Security Index. That being: (a) Income Adequacy, (b) Access to Benefits, (c) Job Flexibility, (d) Income Stability, and (e) Financial Resilience. The foundation is the Walton's eight-dimension QWL model (1973), still cited as the theoretical anchor for nearly every subsequent QWL study, alongside Nanjundeswaraswamy and Sandhya's (2016) synthesis of the field, which works to quantify how frequently each component of QWL appears across decades of published studies. Beyond that, I also quote Sirgy, Efraty, Siegel, and Lee's (2001) need-satisfaction measure, Martel and Dupuis's (2006) organizational-context model, Meibel, Almsafir, Siren, and Alnasser's (2013) driver-frequency review, and Efraty and Sirgy's (1998) earlier empirical test of

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need satisfaction. Together, these sources justify treating the first four dimensions as measurable employment conditions and the fifth, financial resilience, as the household-level outcome those conditions are hypothesized to shape.

Income Adequacy

Income Adequacy is a concept grounded in what Walton (1973) termed “adequate and fair compensation”. Essentially, it is the act of workers having sufficient income in order to cover their essential needs and maintain a standard of living on a day-to-day basis. It is the single most consistently cited QWL component in the literature reviewed by Nanjundeswaraswamy and Sandhya (2016), appearing in 29 of the studies tallied. This is significant for the ESI because perceived adequacy, not raw income, is what the QWL literature links to job satisfaction and reduced financial strain. Empirically, the attribute is best operationalized through self-proposed financial standing as opposed to income independently, since two workers earning equivalent paychecks can experience very different adequacy levels depending on external factors & conditions in their respective lives.

Access to Benefits

Access to benefits corresponds to a distinction the literature draws variably. For instance, Nanjundeswaraswamy and Sandhya (2016) define “Facilities” to include benefits and social security, citing it in eight studies, while Martel and Dupuis (2006) define “Benefits” as its own item separate from “Income” in their organizational-context table. Sirgy et al. (2001) go one step further, isolating “job related health benefits” as an independent dimension from pay entirely, nested instead under health and safety needs rather than economic needs. However, for the purpose of this study, I have defined access to benefits as an individual’s practical opportunity in obtaining financial, health, or social welfare assistance from an employer or organization. Benefits Access earns its place as a standalone dimension because it captures a form of economic protection that income

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alone cannot capture, making it an essential measure of financial security rather than a redundant proxy for pay.

Job Flexibility

For the purpose of this study, job flexibility is defined as employee autonomy to choose factors such as working hours, tasks, and the times at which said tasks are completed. Nanjundeswaraswamy and Sandhya (2016) cite “Autonomy of work”, including flexible time and homework, in eight studies, and Martel and Dupuis (2006) separately list “Work Schedule” and “Flexible Schedules” as organizational-context items. Efraty and Sirgy (1988) locate autonomy under “ego needs” as opposed to “financial security needs”. Yet Mejbél et al. 's (2013) frequency review of 15 studies did not track autonomy or flexibility among its eight drivers at all, a discrepancy worth noting. This suggests that while flexibility is well established in foundational models, it has not been treated as a first priority in every strand of QWL research literature.

Income Stability

Income stability draws on job security, the second most consistently supported construct after income compensation, though the literature labels it in their own diverse methods. Nanjundeswaraswamy and Sandhya (2016) cite it in thirteen to fourteen studies; Sirgy et al. (2001) bundle it with pay under “economic and family needs,” defining it as “feeling secure knowing that one is not likely to get laid off.” Martel and Dupuis (2006) branch the concept into three separate ways within the same table, as “Income security”, “job security”, and “job stability”, suggesting the field has not converged on a single term even within this one study. On the other hand, Mejbél et al. (2013) track “safety and security” drivers in seven of 15 studies, which combines the vast fields of physical workplace safety with employment continuity. For this index, income stability is treated as the predictability of earnings month to month per employee.

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Financial Resilience

Financial resilience remains the one variable with no direct match for its ambiguous nature and title. In this study, financial resilience is defined as a household's capacity to absorb an income shock with little to no after effects. Although, other studies fail to encompass all the various factors that are considered. Such as Sirgy et al. 's (2001) "economic and family needs" category, the most granular treatment of economic needs, stops at pay and job security. Efraty and Sirgy (1988) similarly frame "survival needs" as an economic resource but do not go into depth about emergency funds or repercussions from financial hardships. This gap is not a flaw in the search, but rather a reflection that financial resilience is an overlooked statistic that could change the tide of how we currently envision authentic economic security in the United States.

All together, these five parameters translate literature on job quality into a workable structure for measuring economic well-being: four dimensions capturing what a job provides and one portraying what a household can overcome as a result. The weighing evidence assembled here comes predominantly from a single synthesis, Nanjundeswaraswamy and Sandhya (2016) and their thorough analysis of each dimension accounted for in this study. The next step is to evaluate whether this literature-derived structure holds up in a multidisciplinary approach using real-world data and a computational model in order to determine whether income adequacy, benefits access, job flexibility, income stability, and financial resilience can truly predict the economic security of the American labor force.

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Methodology

Data Sources

This study utilizes data from the 12,934 responses to the 2025 Survey of Household Economics and Decision Making (SHED)² administered by the United States Federal Reserve as the primary dataset for constructing the ESI. This was selected as it contains nationally representative survey data on household finances, access to employer-sponsored benefits, employment conditions, and other relevant indicators to economic success & security. Variables from the dataset were selected based on their alignment with the five dimensions of ESI. To support variable selection, variable weighing, and provide theoretical justification for the model, as previously outlined, evidence from more than eighteen peer-reviewed studies was synthesized throughout the research process in order.

Economic Security Index Construction

To quantitatively evaluate economic well-being, this study developed an ESI composed of five parameters: (a) Income Adequacy, (b) Access to Benefits, (c) Job Flexibility, (d) Income Stability, and (e) Financial Resilience. They were selected based on their consistent identification in over 18 research papers and were measured using responses from the 2025 Survey of Household Economics and Decision Making (see Appendix A). Access to Benefits was calculated as the average of employer-sponsored health insurance (E4_a) and employer-sponsored pension (K21_b). Moreover, job flexibility was formulated by averaging work schedule flexibility (D30), autonomy in selecting work tasks (D28_a), and autonomy in determining how tasks are completed (D28_b). Similarly, Financial Resilience was measured using the average of emergency saving availability (EF1), the ability to cover three months of expenses following income loss

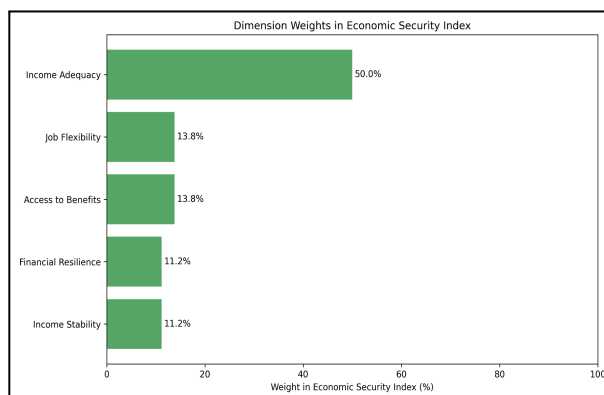
² 2025 Survey of Household Economics and Decision Making (SHED):
<https://www.federalreserve.gov/consumerscommunities/shed.htm>

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(EF2), and respondents' self-assessed financial circumstances (B2). Finally, Income Stability was represented by month-to-month income consistency (I9) and Income Adequacy was measured by I12, which assesses whether participants experienced difficulty paying bills during the previous year.

Seeing as these variables were originally recorded using different response scales, each was standardized to a common 0-100 scale before aggregation. Binary responses were converted to 100 and 0, whilst ordinal answers were propositionally rescaled so higher values represented greater economic security in the long run.

Using data from Nanjundeswaraswamy and Sandhya (2016)'s study on the frequency of QWL variables in other studies, I scaled each dimension to represent a different weighted chunk of the final ESI. As Income Adequacy was referenced in 29 out of the 58 studies, it composes 50% of the total score. Job Flexibility and Access to benefits both compose 11 out of 58 of the total papers, which trickles down into 13.8% each. Finally, Income Stability and Financial Resilience both appear in 8 studies, which reduces to 11.2% of the summative index.



This bar chart displays the individual weighing of the five variables used to determine the ESI. The resulting Economic Security Index ranges from 0 to 100, where higher scores indicate greater overall economic security.

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The final ESI was computed as a weighted sum of the five standardized dimensions:

$$(0.138)\left(\frac{(E4_a + K21_b)}{2}\right) + (0.112)(i9) + (0.50)(i12) \\ + (0.138)\left(\frac{(D30 + D28_b + D28_a)}{3}\right) + (0.112)\left(\frac{(EF1 + EF2 + B2)}{3}\right)$$

Computational Model

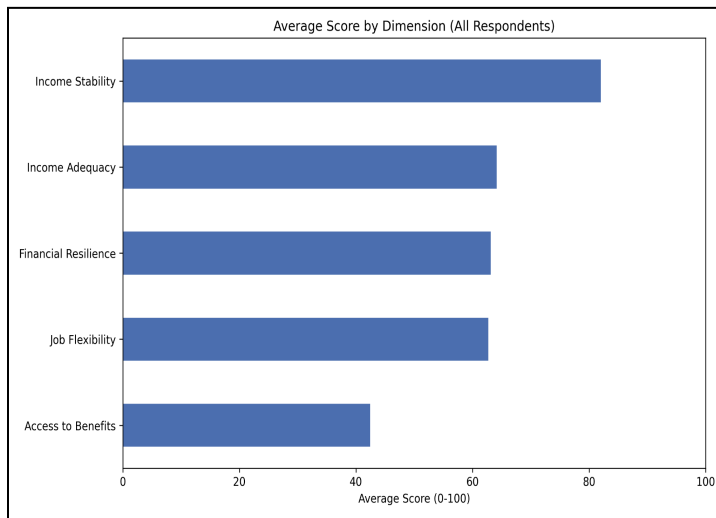
The computational model was developed on Visual Studio Code using Python to automate the analysis of the Economic Security Index (ESI). After cleaning the survey data of unnecessary data beyond the previously identified columns, the model transformed all selected variables to a common 0-100 scale before calculating composite scores for each element. These variables were then combined using the weighted ESI equation to generate the final score for each respondent. Additionally, the model provided a comprehensive analysis through descriptive statistics and generated diagrams illustrating nuanced perspectives not previously highlighted.

The computational approach was selected to ensure that the analysis remained scalable, transparent, and reproducible. Automating the data processing and index calculation minimized potential for manual computation errors while allowing this methodology to be consistently applied across thousands of survey responses. For a further flexible framework, the model can be adapted to incorporate additional variables, alternative weighting schemes, or future data updates.

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Computational Analysis

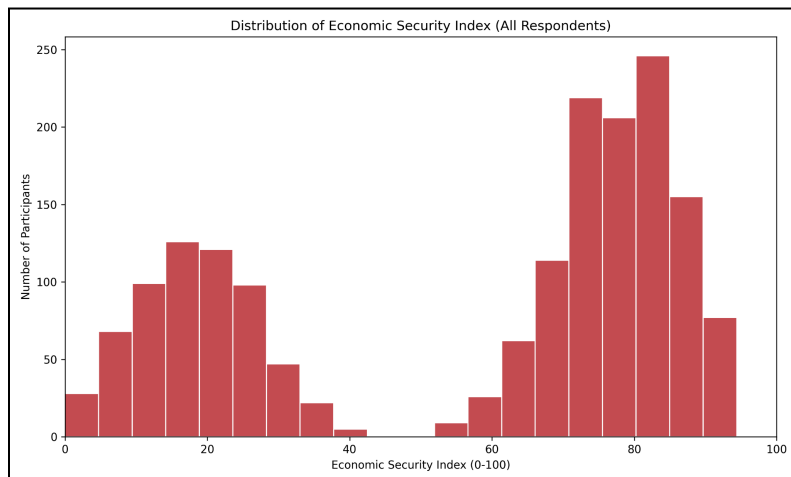
Average Score by Dimension for All Participants



This bar chart illustrates the mean score of each of the five variables for all respondents to the survey. It was collected by first evaluating each dimension on a scale of 0-100 for each participant, after which the scores were averaged within their respective domains. With this output, it is evident that

most participants scored highest in Income Stability. Opposingly, employer-sponsored benefits received the lowest average score, indicating that employer sponsored benefits were less prevalent among this sample population.

Distribution of Economic Security Index for All Participants

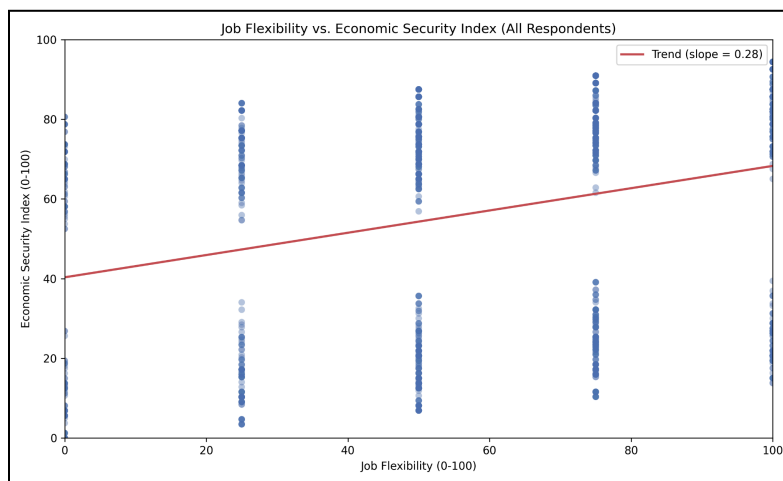


This figure represents the distribution of the final ESI for all respondents to visualize where the average individual lies on this scale. After calculating an ESI score for each individual using the weighted index, the

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scores were grouped into intervals to illustrate the peculiar frequency of the scores. The resulting histogram reveals the highest concentration of respondents lied at the two extremes of the scale, specifically around 20 & 80 on the ESI scale.

Job Flexibility vs. Economic Security Index for All Participants



This scatterplot illustrates the relationship between job flexibility and the ESI for all survey participants. Each point represents an individual participant, with the x-axis representing job flexibility and y-axis demonstrating the ESI.

By plotting these variables together, this diagram evaluates that with a higher range of flexibility, comes a positive correlation with better economic security.

Results & Interpretations

The results of the study demonstrate that security is best understood as a multifaceted construct as opposed to one streamlined financial indicator. By combining Income Adequacy, Access to Benefits, Job Flexibility, Income Stability, and Financial Resilience into a unified index, the prototype revealed meaningful variation in economic well-being across the survey population. Accompanying this, the diagrams outputted by the model displayed the relative performance of each element and their relationship to the overall economic security index, showing that no one single factor fully encompasses the complexities of individual ESI scores. In summary, the verdicts suggest that economic security emerges from the combined influence of multiple

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financial and employment-related characteristics, reinforcing the idea of our society's versatile economic nature.

While this prototype blueprint provides a well-established method for measuring economic well-being, several limitations should be considered when interpreting the results. For instance, the available studies and survey data do not distinguish well between income stability and financial resilience, as there is some conceptual overlap. To fix this issue, separate variables were selected to represent each construct in a method that eliminated most of the conceptual overlap. This is further elaborated on in Appendix A. Additionally, the analysis relies on self-reported survey responses, which may be subject to reporting response bias. Finally, the index does not fully account for various factors that could influence economic security, including demographic disparities and mental health. To further refine and validate the ESI, future research could incorporate these factors with their respective weighting.

Conclusion

Developing an effective index of economic security assisted in addressing the question of how to quantify economic security in a time of precarious market circumstances and more specifically, how do specific dimensions affect the outcome of financial stability. By integrating evidence from data of the 2025 Survey of Household Economics and Decision Making with existing literature, this research developed a computational model capable of constructing an Economic Security Index based on five key dimensions: (a) Income Adequacy, (b) Access to Benefits, (c) Job Flexibility, (d) Income Stability, and (e) Financial Resilience. As opposed to relying on a single indicator such as income, the suggested outline harvests the intersection of various variables to give a comprehensive view on the matter.

Although this form of measurement offers a robust blueprint for evaluating economic well-being, the study was subject to several constraints. For instance, the results of the data source used to create the model relies on self-reported survey data, which may show bias or dishonest

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responses. Future research could expand upon these findings by incorporating longitudinal data, regional cost-of-living adjustments, demographic disparities, or additional indicators of financial well-being to account for omitted variables in order to further refine the model. Despite these limitations, the proposed methodology establishes a scalable foundation for measuring economic security. With the conclusion of this study, policymakers, businesses, and government institutions can now support the United States labor force with an extensive and empirical economic security index.

AI use transparency statement

The python code used to establish the final models was partially developed using Claude Sonnet 5.0. It was additionally used to check for any spelling, grammar, or sentence structure errors throughout this paper. The author independently revised and edited the project as needed, and takes full responsibility for the conclusions presented in this paper.

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Appendix A

Variable Standardization

Variable	Description	Original Scale	Standardized Scale
E4_a	Do you have health insurance from your employer?	Yes/No	100/0
K21_b	Do you have a pension w/ defined benefit from your employer?	Yes/No	100/0
D30	Do you have normal hours or does it vary?	Schedule varies at my request/ Normally work same hours/ Schedule varies based on employer needs	100/50/0
D28_a	Do you choose the tasks you work on at work?	Yes/No/Unsure please explain	100/0/Omitted
D28_b	How often do you choose how to complete tasks at work?	Always/Often/Sometimes/ Rarely/Never	100/75/50/25/0
EF1	Do you have emergency funds to cover expenses for 3 months in case of sickness, job loss, economic downturn, or other emergencies?	Yes/No	100/0

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EF2	If you lost your main source of income, could you cover expenses by borrowing money, using savings, or buying assets?	Yes/No	100/0
B2	How are you managing financially these days?	Living comfortably/ Doing okay/ Just getting by/ Finding it difficult to get by	100/67/33/0
I9	Is your income stable from month to month?	Yes/No	100/0
I12	Have you and your partner struggled to pay your bills in the past 12 months?	Yes/No	100/0